

Conducting Wi-Fi Reconnaissance

Using the Aircrack-ng Suite

Sahithyan K. (230557T)

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1. Introduction

Wi-Fi reconnaissance is the process of collecting information about nearby wireless networks before conducting security assessments or penetration testing activities. It helps identify wireless access points, security configurations, connected devices, and possible vulnerabilities within a wireless environment.

In this assignment, the Aircrack-ng suite was used to perform Wi-Fi reconnaissance. Aircrack-ng is a collection of tools used for wireless network monitoring, packet capturing, and security analysis.

1.1 Tools Used

- **airmon-ng**: enables monitor mode on the wireless adapter.
- **airodump-ng**: scans nearby wireless networks and captures packets.
- **aireplay-ng**: performs deauthentication attacks for capturing WPA/WPA2 handshake packets.

The reconnaissance process focused on gathering information such as SSIDs, BSSIDs, channels, encryption types, and signal strengths of nearby wireless networks.

2. Preparation

Before starting, several preparations were made to ensure the tools and wireless adapter worked correctly.

The reconnaissance was initially planned on a MacBook, but the built-in Wi-Fi adapter did not support monitor mode or packet injection, which are required by **airodump-ng** and **aireplay-ng**. A Lenovo Vostro 15 running Ubuntu 24.04 was used instead. It has a Realtek RTL8852BE PCIe 802.11ax wireless adapter with full support for monitor mode and the Aircrack-ng suite.

2.1 Background

2.1.1 802.11 Wireless Communication

Wi-Fi devices communicate using the IEEE 802.11 family of standards over shared radio frequencies. All frames broadcast over the air are visible to any nearby device operating in *monitor mode*, which allows passive capture without association to any network. 802.11 defines three frame types: management frames (beacons, probe requests, authentication, deauthentication), control frames, and data frames. Management frames in WPA and WPA2 networks are sent in plaintext and carry no authentication, which is the root cause of the deauthentication vulnerability exploited later in this assignment.

2.1.2 WPA, WPA2, and WPA3

WPA (Wi-Fi Protected Access, 2003) was introduced as an emergency replacement for the broken WEP standard. It uses the TKIP cipher, which wraps RC4 with per-packet key mixing. TKIP is now considered cryptographically weak and is deprecated.

WPA2 (IEEE 802.11i, 2004) replaced TKIP with the AES-based CCMP cipher, which is significantly stronger. Two authentication modes exist: Personal (PSK, a shared passphrase) and Enterprise (802.1X with a RADIUS server). WPA2-Personal is the most common deployment

and is the target of offline passphrase-cracking attacks once the four-way handshake is captured. **WPA3** (2018) introduced Simultaneous Authentication of Equals (SAE). SAE replaces the PSK handshake with a Diffie-Hellman-based exchange, providing forward secrecy and resistance to offline dictionary attacks. WPA3 also mandates Protected Management Frames (PMF/802.11w), which prevents the deauthentication attack described in Section 3.3.

Standard	Year	Cipher	Key Vulnerability	Status
WEP	1997	RC4 (static key)	Key reuse; crackable in minutes	Broken
WPA	2003	TKIP (RC4 wrapped)	Weak MIC; offline dictionary attacks	Deprecated
WPA2	2004	AES-CCMP	Offline handshake cracking (PSK mode)	Widely used
WPA3	2018	AES-GCMP + SAE	No known practical attacks	Recommended

2.1.3 The WPA/WPA2 Four-Way Handshake

When a client connects to a WPA2 network, a four-message exchange derives the session key (PTK) from the PSK and two random nonces from the access point and client. No plaintext password is transmitted, but the handshake contains enough material for offline attacks: a candidate passphrase is used to recompute the MIC (message integrity code), and if it matches the captured value, the passphrase is correct. Capturing this handshake is the primary goal of the targeted capture and deauthentication steps below.

2.2 Installing Aircrack-ng

The Aircrack-ng suite was installed on a Linux system using the package manager.

```
sudo apt install aircrack-ng
```

2.3 Checking Wireless Adapter Compatibility

Monitor mode and packet injection support were required. The wireless interface was verified using:

```
iwconfig
```

The detected wireless interface was `wlp2s0`.

2.4 Disabling Background Services

To prevent conflicts with the wireless interface, potentially interfering processes were stopped:

```
sudo airmon-ng check kill
```

The following processes were identified and killed:

PID	Name
1236	wpa_supplicant
1323558	avahi-daemon
1323562	avahi-daemon

2.5 Enabling Monitor Mode

To capture wireless traffic, monitor mode was enabled:

```
sudo airmon-ng start wlp2s0
```

After enabling monitor mode, the interface was renamed to `wlp2s0mon`.

3. Activities, Results, and Interpretation

3.1 Passive Network Scanning

Passive scanning is the first step in any wireless assessment: the adapter listens without transmitting, so no packets are sent to any network and no interaction occurs. This phase identifies potential targets (BSSIDs, channels, encryption types, and connected clients) before any active technique is attempted.

Nearby wireless networks were scanned using `airodump-ng`:

```
sudo airodump-ng wlp2s0mon
```

This command displayed information about nearby access points and connected clients, including SSID, BSSID, channel, encryption type, signal strength, and associated clients. Six scanning sessions were conducted on 2026-05-16 (12:27–12:54), totalling approximately 27 minutes of passive capture. A total of **39 unique access points** were detected.

A representative selection of the detected networks is shown below, ordered by signal strength:

SSID	BSSID	Ch.	Encryption	Signal
Thavanathan's iPhone	A6:8E:DC:82:EB:7B	6	WPA2	−47 dBm
Galaxy M0298c8	22:7C:A4:7D:D9:64	6	WPA2 / Open	−64 dBm
Boss	9E:02:9F:A9:99:85	1	WPA2	−66 dBm
Redmi A1+	BA:3D:EB:6D:12:5C	6	Open	−71 dBm
Dialog 4G	7C:11:CB:07:F3:03	6	WPA2	−76 dBm
Dialog 4G 777	98:A9:42:17:FE:C1	7	WPA2	−77 dBm
Dialog 4G 379	D8:42:F7:E2:21:7B	10	WPA2	−79 dBm
Redmi A5	4E:A8:A8:1D:FA:AF	1	WPA3 / WPA2	−84 dBm
EZVIZ_BA8983061	BA:4D:43:17:49:3D	3	WPA2 / WPA	−85 dBm
SLT-4G-EAFC	0C:8F:FF:28:14:7C	6	WPA2 / WPA	−91 dBm

Observations:

- **32 networks** used WPA2 encryption; **1 network** (Redmi A5) used WPA3.
- **4 networks** were fully open (no encryption), including the “Redmi A1+” hotspot.
- **7 networks** broadcast with a hidden SSID (empty ESSID field).
- Several networks advertised both WPA2 and the older WPA standard simultaneously, indicating mixed-mode or legacy configurations.
- Multiple mobile hotspots were visible, including devices named “Thavanathan’s iPhone”, “Galaxy M0298c8”, and “TimBaTTuK’s S25 Ultra”.
- IoT and surveillance devices were identifiable by their SSIDs: EZVIZ_ (security camera),

DDPAI_ and 70mai_ (dash cameras), and three WIFIBRG_ bridge units.

- Vehicle-mounted access points were present: “Vezel-Dash-Cam” and “In_Vehicle_Wifi”.
- Up to 3 associated client devices were observed around a single access point during capture, making client tracking possible.

Interpretation:

The scan results illustrate how much a passive observer can learn without authenticating to any network. Four open networks expose all transmitted traffic in plaintext, leaving users vulnerable to credential theft and session hijacking. The 32 WPA2 networks provide reasonable protection with strong passphrases, but several ISP and IoT devices broadcast mixed WPA/WPA2 configurations, which reintroduce the weaker TKIP cipher and can be exploited. Only one network used WPA3, underscoring its slow real-world adoption. The seven hidden SSIDs offer no practical security benefit: access points still broadcast beacon frames and were captured trivially by `airodump-ng`. IoT devices (cameras, dash cameras, bridge units) advertise distinctive SSIDs and typically run outdated firmware, making them attractive entry points for network intrusion.

3.2 Targeted Packet Capture

A specific wireless network was selected for deeper packet capture:

```
sudo airodump-ng -c 6 --bssid A6:8E:DC:82:EB:7B -w capture wlp2s0mon
```

- `-c`: specifies the channel number
- `--bssid`: specifies the target MAC address
- `-w`: writes captured packets to a file

The capture session ran from **12:56:33** to **12:57:12** (approximately 39 seconds). The following access point and client data were recorded:

SSID	BSSID	Ch.	Encryption	Signal	Beacons	Data
Thavanathan's iPhone	A6:8E:DC:82:EB:7B	6	WPA2 (CCMP/PSK)	−78 dBm	322	1407

Two client devices were observed associated with this access point:

Client MAC	Signal	Packets	Note
24:EB:16:56:E1:CC	−76 dBm	3124	Directed deauth target (succeeded)
14:5A:FC:81:7E:1D	−37 dBm	1134	First deauth attempt (channel mismatch)

Interpretation:

Locking `airodump-ng` to a single channel and BSSID focuses all captured frames on one target, enabling reliable client enumeration in a short time. The 1,407 data frames captured in under 40 seconds show that even brief sessions yield enough material for further analysis.

3.3 Deauthentication Attack

A deauthentication attack was performed against an isolated private Wi-Fi access point (a personal mobile hotspot) to force connected clients to reconnect and generate WPA handshake packets. The target was selected because it was under personal control and no third-party devices were

affected.

3.3.1 Broadcast Deauthentication

An initial broadcast deauthentication attack was sent to all clients associated with the target access point:

```
sudo aireplay-ng -0 5 -a A6:8E:DC:82:EB:7B wlp2s0mon
```

This sent five deauthentication frames to the broadcast address, requesting all connected clients to disconnect from the access point.

3.3.2 Directed Deauthentication

A directed attack targeting a specific client was then attempted. An initial attempt failed because the monitor interface was locked to channel 13 while the access point operated on channel 6:

```
sudo aireplay-ng -0 5 -a A6:8E:DC:82:EB:7B -c 14:5A:FC:81:7E:1D  
wlp2s0mon
```

Running `airmon-ng check kill` resolved the channel lock issue. A subsequent directed deauthentication against client 24:EB:16:56:E1:CC succeeded:

```
sudo aireplay-ng -0 5 -a A6:8E:DC:82:EB:7B -c 24:EB:16:56:E1:CC  
wlp2s0mon
```

`aireplay-ng` reported 63 out of 67 ACKs for the sent frames, confirming the client received and acted upon the deauthentication packets.

A continuous attack (-0 0) was then run for approximately five minutes to observe sustained deauthentication behaviour. ACK counts varied across packets, indicating intermittent reconnection attempts by the client.

3.3.3 Fake Authentication Attempt

A fake authentication request was also sent to the access point to test association behaviour:

```
sudo aireplay-ng -1 0 -a A6:8E:DC:82:EB:7B -c 24:EB:16:56:E1:CC  
wlp2s0mon
```

The open-system authentication succeeded, but the subsequent association request was denied with status code 12 (*wrong ESSID or WPA*), confirming that WPA2 protection was enforced and association without valid credentials was not possible.

Interpretation:

The attack exposed a fundamental weakness in 802.11: management frames such as deauthentication are unauthenticated, so any station can forge and send them without network credentials. The high ACK rates (63 out of 67) confirm the client received and acted on the frames. WPA3 networks with Management Frame Protection (MFP) are immune because their management frames are cryptographically signed. Since almost no detected networks used WPA3, this technique would work against nearly all observed access points.

3.4 WPA Handshake Capture

During reconnection triggered by the deauthentication attack, WPA handshake packets were captured successfully. The handshake notification appeared in the top-right corner of the `airodump-ng` window:

```
WPA Handshake : A6:8E:DC:82:EB:7B
```

Interpretation:

Capturing the four-way WPA handshake is the prerequisite for offline passphrase cracking. Once captured, an attacker can run dictionary or brute-force attacks using tools such as `aircrack-ng` or `hashcat` with no further interaction with the network. A weak passphrase can therefore be recovered long after the capture. The collected information could be used for unauthorised network access, password cracking, traffic monitoring, device tracking, and denial-of-service attacks.

3.5 Passphrase Cracking Attempt

To illustrate the offline cracking phase, `aircrack-ng` was run against the captured handshake file using the `rockyou.txt` wordlist, one of the most widely used password dictionaries in security assessments:

```
aircrack-ng -w /usr/share/wordlists/rockyou.txt capture-01.cap
```

The attack ran for approximately 20 minutes running **14,344,391** candidate passphrases at a sustained rate of approximately **11,780 keys per second**. The correct passphrase was **not found** within the rockyou list.

The failed dictionary attack shows that a strong passphrase is an effective defence against offline attacks, even after the four-way handshake is captured. Recovering such a passphrase would require larger wordlists, rule-based mutations, or full brute-force, each increasing time and computational cost by orders of magnitude.

4. Summary and Conclusion

This assignment demonstrated the process of conducting Wi-Fi reconnaissance using the Aircrack-ng suite. The tools `airmon-ng`, `airodump-ng`, and `aireplay-ng` were used to scan nearby wireless networks, gather information, and capture WPA/WPA2 handshake packets.

Across six scanning sessions conducted on 2026-05-16 (12:27–12:54), 39 unique access points were identified. Their SSIDs, BSSIDs, channels, encryption configurations, and signal strengths were collected passively without authenticating to any network. The activity also demonstrated how deauthentication attacks can force clients to reconnect and generate handshake packets for offline analysis.

The findings highlighted several real-world wireless security concerns: the presence of open hotspots, wide deployment of legacy mixed WPA/WPA2 configurations, near-absent WPA3 adoption, inadequately obscured IoT devices, and the fundamental insecurity of unauthenticated 802.11 management frames.

4.1 Recommendations

The following measures can improve wireless security:

-
- Use WPA2 or WPA3 encryption
 - Avoid open wireless networks
 - Use strong, unique passwords
 - Disable WPS functionality
 - Regularly update router firmware
 - Monitor connected devices regularly

In conclusion, Wi-Fi reconnaissance is an important part of wireless security assessment because it helps identify vulnerabilities and security weaknesses before attackers can exploit them.